PH 724 Data Analysis & Scientific Communication Incorporating Use of Generative AI

Part 7: Introduction to the Gaussian Process

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What is a Gaussian process (GP)

- **Intuition:** a series of random variables are connected to each other, forming a random curve.
- Any finite set of points on the random curve has a joint multivariate normal distribution.
- Specifically, in math, we write a random curve $Y(t) \sim \text{GP}\{\mu(t), K(t,t')\}$ if, for any finite collection of times $t_1, \ldots, t_m$,

$$\begin{pmatrix} Y(t_1) \\ \vdots \\ Y(t_m) \end{pmatrix} \sim N\left(\begin{pmatrix} \mu(t_1) \\ \vdots \\ \mu(t_m) \end{pmatrix}, [K(t_j, t_k)]_{j,k=1}^{m} \right).$$

Two ingredients: mean and covariance functions

- Mean function: $\mu(t) = E\{Y(t)\}$
  - Describes the expected trajectory
- Covariance function: $K(t,t') = \text{Cov}\{Y(t), Y(t')\}$
  - Describes how the covariance varies across times.

The covariance function matters

Consider two possible covariance structures:

- **Smooth covariance:** nearby times tend to have similar deviations; sample paths are smooth.
- **Brownian-motion covariance:** deviations accumulate over time; paths are continuous but rough.

Sample paths of a GP with a smooth covariance function

```
library(ggplot2)
library(MASS)
set.seed(724)

# Number of simulated curves
n_curves <- 30

t <- seq(0, 1, length.out = 100)
mu <- 30 * t^2 / (0.25 + t^2)
```

```r
K <- outer(t, t, function(a, b)
  6^2 * exp(-(a-b)^2 / (2*0.18^2)))

# Generate more simulated trajectories
paths <- MASS::mvrnorm(n_curves, mu, K + diag(1e-7, length(t)))

# Create data frame
d <- data.frame(
  time = rep(t, n_curves + 1),
  height = c(mu, as.vector(t(paths))),
  trajectory = rep(
    c("Mean", paste("Possible plant", 1:n_curves)),
    each = length(t)
  )
)

# Plot
ggplot(d, aes(time, height,
              group = trajectory,
              color = trajectory == "Mean")) +
  geom_line(aes(alpha = trajectory == "Mean"), linewidth = 0.8) +
  scale_color_manual(
    values = c("TRUE" = "#202020", "FALSE" = "#7a9fbd"),
    guide = "none"
  ) +
  scale_alpha_manual(
    values = c("TRUE" = 1, "FALSE" = 0.4),
    guide = "none"
  ) +
  labs(
    x = "Scaled time",
    y = "Height (illustrative units)",
    subtitle = "Illustration only: 30 simulated trajectories from a smooth GP"
  ) +
  theme_minimal(base_size = 16)
```

Brownian motion as a Gaussian process

For standard Brownian motion $W(t)$, starting at $W(0) = 0$,

$$E\{W(t)\} = 0, \qquad \mathrm{Cov}\{W(t), W(t')\} = \min(t, t').$$

- The start point of each trajectory is fixed at 0. - Variability increases with elapsed time: $\mathrm{Var}\{W(t)\} = t$. - Brownian paths are “bumpy”.

Sample paths of a standard Brownian motion

```r
library(ggplot2)
set.seed(724)

# Number of simulated Brownian-motion paths
n_curves <- 30
```

```r
tb <- seq(0, 1, length.out = 150)

# Generate Brownian-motion paths
B <- replicate(
  n_curves,
  c(0, cumsum(rnorm(length(tb) - 1, sd = sqrt(diff(tb)[1]))))
)

# Create data frame
db <- data.frame(
  time = rep(tb, n_curves),
  deviation = as.vector(B),
  plant = rep(1:n_curves, each = length(tb))
)

# Plot
ggplot(db, aes(time, deviation, group = plant)) +
  geom_line(
    color = "#7a9fbd",
    linewidth = 0.8,
    alpha = 0.4
  ) +
  geom_hline(
    yintercept = 0,
    color = "#202020",
    linewidth = 1,
    linetype = "dashed"
  ) +
  labs(
    x = "Elapsed time",
    y = "Deviation",
    subtitle = "Illustration only: 30 simulated standard Brownian-motion paths"
  ) +
  theme_minimal(base_size = 16)
```

Real-world applications of GPs

- One of the most influential mathematical finance models is the Black–Scholes model, developed in 1973. It assumes that the stock price follows a geometric Brownian motion (whose logarithm is a GP). Black-Scholes uses this stochastic model primarily for option pricing, not for reliably predicting tomorrow's stock price.
- Google has used GP-based Bayesian optimization to efficiently tune machine-learning models and other complex systems. GPs help identify promising configurations while reducing the number of expensive evaluations. See https://deepmind.google/research/publications/108347/.
- GPs are widely used to model spatially varying quantities, such as environmental pollution levels and disease prevalence.
  - For example, the Malaria Atlas Project uses GP-based spatial modeling to estimate malaria prevalence in regions with limited observational data. Its broader modeling efforts also estimate malaria incidence and mortality, producing high-resolution maps that support public health research and decision-making.

Textbook's growth-curve model

For plant i in strain $s(i)$, the textbook considers a model of the form

$$Y_i(t) = \beta_{s(i)} h(t) + \sigma_1 W_0(t) + \sigma_2 W_i(t) + \varepsilon_i(t),$$

where $h(t) = t^2/(\tau^2 + t^2)$ is a candidate mean-curve shape.

- $\beta_{s(i)} h(t)$: strain-specific mean growth curve.
- $W_0(t)$: **shared** standard Brownian-motion process.
- $W_i(t)$: **plant-specific** standard Brownian-motion process.
- $\varepsilon_i(t)$: measurement error.

Simulated growth trajectories under the textbook's GP model

```
library(ggplot2)
set.seed(724)

# Number of simulated plants in each strain
n_cis <- 40
n_108 <- 30
n_plants <- n_cis + n_108

# Time since estimated emergence (days)
t <- seq(0, 40, length.out = 150)
n_time <- length(t)

# Illustrative model parameters (not fitted estimates)
beta <- c(cis = 28, `108` = 33)  # Mean asymptotic heights (mm)
tau <- 12                        # Growth time-scale parameter
sigma1 <- 0.30                   # Shared Brownian-motion variation
sigma2 <- 0.65                   # Plant-specific Brownian variation
sigma_e <- 0.50                  # Measurement-error SD

# Textbook mean growth function
h <- function(t, tau) {
  t^2 / (tau^2 + t^2)
}

# Standard Brownian motion on a specified time grid
brownian_motion <- function(t) {
  c(0, cumsum(rnorm(
    length(t) - 1,
    mean = 0,
    sd = sqrt(diff(t))
  )))
}

# Strain membership
strain <- c(rep("cis", n_cis), rep("108", n_108))

# One Brownian path shared by all plants
W0 <- brownian_motion(t)
```

```r
# Independent Brownian paths for individual plants
Wi <- replicate(n_plants, brownian_motion(t))

# Simulated trajectories
growth <- do.call(
  rbind,
  lapply(seq_len(n_plants), function(i) {

    mean_height <- beta[strain[i]] * h(t, tau)

    latent_height <- mean_height +
      sigma1 * W0 +
      sigma2 * Wi[, i]

    observed_height <- latent_height +
      rnorm(n_time, sd = sigma_e)

    data.frame(
      time = t,
      plant = i,
      strain = strain[i],
      mean_height = mean_height,
      latent_height = latent_height,
      observed_height = observed_height
    )
  })
)

# Mean curves for the two strains
mean_curves <- expand.grid(
  time = t,
  strain = c("cis", "108")
)

mean_curves$height <- beta[mean_curves$strain] *
  h(mean_curves$time, tau)

# Plot simulated latent trajectories and mean curves
ggplot(growth, aes(time, latent_height, group = plant)) +
  geom_line(
    color = "#7a9fbd",
    linewidth = 0.6,
    alpha = 0.35
  ) +
  geom_line(
    data = mean_curves,
    aes(x = time, y = height),
    inherit.aes = FALSE,
    color = "#b4443b",
    linewidth = 1.5
  ) +
  facet_wrap(~ strain) +
  labs(
```

```
    x = "Time since estimated emergence (days)",
    y = "Plant height (mm)",
    subtitle = paste(
      "Simulated growth trajectories under the textbook's",
      "Gaussian-process model"
    ),
    caption = paste(
      "Blue: individual latent trajectories;",
      "red: strain-specific mean growth curves.",
      "Parameters are illustrative."
    )
  ) +
  theme_minimal(base_size = 16)
```